When strong players lead: Endogenous timing in contests with future cost uncertainty

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Abstract

This paper studies endogenous timing in asymmetric Tullock contests when future effort costs are uncertain. Two players with different prize valuations choose whether to exert effort early or late. While one player knows future costs perfectly, the other knows only their distribution. We identify conditions under which the standard timing result is reversed, i.e., the strong player moves first despite conventional wisdom suggesting weak players should lead. This reversal occurs when cost uncertainty is resolved favorably for late movers, but only the weak player is aware of this. The study also examines optimal contest design, showing that strategic information disclosure can generate higher total effort than price discrimination. By fully informing the strong player about unfavorable future costs while maintaining the weak player's optimism, a contest designer can induce effort-maximizing sequential play.

Keywords: Contests; Endogenous timing; Asymmetric information; Cost uncertainty; Information design

JEL Classification: C72, D44, D82

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1 Introduction

On several occasions, individuals or groups of individuals compete with each other to win a prize. They do so by exerting costly efforts, e.g., monetary expenditures or time spending. The prize might include monopoly rents, favorable policies, sports awards, etc.[1] It is common in the relevant literature that the individuals participating in such contests exert efforts simultaneously, i.e., each player makes an effort without knowing the level of effort chosen by the other player. However, there are cases in which the contestants might exert efforts sequentially, i.e., a contestant might exert effort after she has observed the effort exerted by her opponent. For instance, an interest group might choose how much money to contribute to its favorite candidate during election campaigning before the opposing group does.

Although several studies endogenize the timing of the players' efforts in a contest, i.e., each player can choose whether she wants to make an effort early and become the leader or late and become the follower,[2] to our knowledge, no study considers uncertainty about the future state of the world, namely, the effort cost when the follower chooses how much effort to submit. For example, if a lobby decides to wait for the opponent to contribute first, when the time for its contribution arrives, the lobbying effort cost might differ from the one the opponent lobby pays now. Similarly, in an innovation contest, the investment made by the leader innovator might affect the price the follower innovator will have to pay.[3]

A well-established result in the endogenous timing literature is that the weak player, i.e., the one who values the prize lower, moves early while the strong player follows (Baik & Shogren, 1992; Leininger, 1993). This occurs because both players benefit from this sequence: the weak leader commits to moderate effort, inducing a less aggressive response from the strong follower compared to simultaneous play. However, this result has been derived under complete information.

In this paper, we study a Tullock contest with two players who are *ex ante* asymmetric

[1] For surveys on the contest theory literature, see Konrad (2009) and Beviá & Corchón (2024).

[2] For studies in endogenous timing in contests, see Baik & Shogren (1992), Linster (1993), Leininger (1993), Morgan (2003), Konrad & Leininger (2007), Hoffmann & Rota-Graziosi (2012), Baik & Lee (2013), Baik et al. (2022), Protopappas (2023), Baik & Lee (2024) and Gao et al. (2025).

[3] In this study, we do not consider dependency between the present effort and the future cost but, instead, the future cost can be either lower or higher regardless of the leader's effort level.

in prize valuations and have the opportunity to exert effort early or late, i.e., to be the first or the second mover. This implies that the contestant who moves late can observe the first mover's effort before she chooses how much effort to put (Dixit, 1987). While we assume there is perfect and symmetric information about the two players' prize valuations and the present effort cost, there is asymmetric information about the future cost. Particularly, one of the players (informed) knows the exact level of the future effort cost, but the other (uninformed) knows only the distribution of it. Therefore, when the two players decide on the timing of their moves, the uninformed player—which can be either the one with the high prize valuation (strong) or the one with the low prize valuation (weak)—does not know whether the effort cost will be lower or higher in the case she decides to make her move late. Instead, she only knows whether she is the strong or the weak player, as well as the present effort cost.

We study both cases of information asymmetry and find the two players' equilibrium efforts. Our main finding is that under certain conditions regarding cost uncertainty and players' beliefs, the conventional result can be reversed: the strong player may choose to move first while the weak player follows. This reversal occurs when the strong player faces sufficient uncertainty about future costs while the weak player has superior information suggesting favorable future conditions. For instance, in the case of an uninformed strong player and an informed weak player, if the present cost is high and the future one is low, a timing equilibrium with this reversed order emerges when the weak player is sufficiently strong and the strong player is not too optimistic about future costs. The intuition is straightforward: the strong player, believing favorable future costs are unlikely, secures the first-mover advantage, while the weak player, knowing costs will fall, prefers to wait. However, if the weak player is too weak, she is preempted entirely. In the case of an informed strong player and an uninformed weak player, a different mechanism drives the reversal. Here, an equilibrium in which the strong player moves first arises when both present and future costs are high, but the weak player is sufficiently optimistic about future cost reductions. The informed strong player, knowing costs will remain high, exploits the first-mover advantage, while the uninformed weak player delays in hopes of lower costs that never materialize.

In the last part of the study, we consider a fully informed contest designer who maxi-

mizes her expected payoff and can manipulate the information she discloses regarding the future cost. We compare the total effort of the contest with equal costs in the two stages and information manipulation to the contest in Protopappas (2023), where the contest designer charges the first mover a lower price, and there is full information. We show that it is optimal for the contest designer to fully disclose information to the strong player and let the weak player believe that the future cost will be low with a high probability rather than charging the first mover a lower price. By doing so, the contest designer can achieve a greater payoff by inducing the strong player to move first and the weak player to follow, while in Protopappas (2023), the best she can achieve is a simultaneous move by the two players.

1.1 Link to the literature

The paper contributes to the contest theory literature by combining two topics in this area: endogenous timing of moves and asymmetric information in Tullock contests. Beginning with discussing the former, the study of Dixit (1987) considers a two-player contest where the players can strategically pre-commit to their levels of effort. It finds that if the favorite player has the opportunity to move early, she commits to a level of effort greater than the Nash equilibrium.[4]

Introducing endogenous timing, Baik & Shogren (1992) and Leininger (1993) consider a Tullock contest with full information where the players can choose their order of moves before the level of effort. They demonstrate that, in equilibrium, the sequence of moves is the one with the player who values the prize lower (weak player) exerting effort early and the one who values the prize higher (strong player) following. This sequence of moves arises in equilibrium because it benefits both players, as the weak player—by moving early—chooses a moderate level of effort. Then, the strong player responds with a greater effort but still lower than the one she would choose if she moved early. Therefore, this order of moves makes participation in the contest "cheaper" for the players compared to the case where they move in an inverse order or even simultaneously. In a later study, Hoffmann & Rota-Graziosi (2012) consider the case with an endogenous prize and show

[4]See also Baye & Shin (1999) for an elaboration on the second-order conditions of the symmetric case in Dixit (1987).

that players may decide to play simultaneously in the sub-game perfect equilibrium. Another study that examines endogenous timing in contests is Morgan (2003), which demonstrates that—under specific assumptions (Serena, 2017)—sequential play by the two players is the only equilibrium outcome. However, in that study, the players are assumed to be *ex ante* identical, different from our study, where the players are *ex ante* asymmetric both in strengths and information possession.

Departing from the settings with symmetric information, Fu (2006) extends the above studies, introducing asymmetric information about the value of the prize. Particularly, when the players decide on the timing of their moves, they do not know the value of the prize, but when the contest begins, the prize is realized, and one of them is informed of it while the other is not. The study's main result is that, in the timing equilibrium, either the two players move simultaneously or the uninformed player leads and the informed one follows. In our model, in contrast to Fu (2006) where the players differ only in the level of information they possess, we assume that the players are asymmetric not only in information possession but also in their strengths, i.e., their prize valuations. Moreover, while in Fu (2006) the level of the early effort by the informed player makes the model a signaling game with that player signaling her type to the uninformed one, in our model, it is the timing choice that might signal the level of the future cost. In particular, if the uninformed player chooses to exert effort early and the informed player follows, the former may infer the future cost level by observing the timing of her opponent's move. On the other hand, an early move by the informed player does not affect her opponent's behavior, as all the information becomes common knowledge before the latter chooses how much effort to put.

Some other studies close to this one are Baik & Shogren (1994) and Protopappas (2023), which also explore endogenous timing in Tullock contests. Nevertheless, their main focus is on how an effort reimbursement or a first-mover's cost advantage in favor of the contest designer might affect the players' move choices. In particular, in the former study, the strong contestant is the one who receives a greater reimbursement in case of winning the contest, and in the latter, a contest designer offers a lower effort price to the player who chooses to move early. In the equilibria of both these games, players either move simultaneously or the weak player moves early and the strong player

follows. Finally, in two recent studies, Baik et al. (2022) and Baik & Lee (2024) consider endogenous timing in a Tullock contest with three players but do not find any equilibria in which the strong player makes an effort early.

The remainder of the paper is structured as follows. In Section 2, we present an overview of the model. In Section 3, we analyze all the possible subgames. In Section 4, we study the players' expected payoffs before the timing decisions have been made, and in Section 5, we introduce endogenous timing in the players' efforts. Lastly, in Section 6, we study the model from a contest designer's perspective while Section 7 concludes the paper. All the proofs are included in the Appendix.

2 Outline of the model

Consider a Tullock contest with two players competing with each other to win a prize. The strong [weak] player's prize valuation is V [v] with $V > v > 0$, which is common knowledge. The two players aim at maximizing their expected payoffs and, therefore, exert costly efforts to increase their probability of winning the prize. Making use of the popular lottery contest success function proposed by Tullock (1980) and axiomatized by Skaperdas (1996) and Clark & Riis (1998), we assume that each player's probability of winning the prize equals the ratio of this player's effort to the sum of efforts. Thus, if player i makes effort x_i her probability of winning is[5]

$$p_i(x_i, x_j) = \begin{cases} \dfrac{x_i}{x_i + x_j}, & \text{if } \max\{x_i, x_j\} > 0, \quad i \neq j \\ \dfrac{1}{2}, & \text{otherwise.} \end{cases}$$

Before the contest begins, the present effort cost (marginal cost), c_1, is common knowledge and equal to $c_L = k$ or $c_H = 1$, where $0 \leq k \leq 1$.[6] While one player (the *informed* player) has perfect information about the future effort cost, the other player (the *uninformed* player) only knows that the effort cost at stage 2, c_2, will be either equal

[5]This contest success function is a special case of $p_i(x_i, x_j) = \frac{x_i^r}{x_i^r + x_j^r}$ with $r = 1$. For $r \neq 1$, because of the sequential setting, the analysis becomes much more complicated and, inevitably, intractable.

[6]The assumption of binary costs has been made for tractability and does not affect the paper's main message. The normalization $c_H = 1$ is without loss of generality.

to k with probability q or equal to 1 with probability $1-q$, where $0 \leq q \leq 1$. In other words, the future cost can either remain the same or change, depending on the present cost.

The timing of the game is as follows. At stage zero, the two players simultaneously choose stage 1 (early) or stage 2 (late) as the time to make their efforts. Importantly, at this timing decision stage, the informed player knows the exact value of c_2 while the uninformed player knows only that $c_2 = k$ with probability q and $c_2 = 1$ with probability $1-q$. Both players know each other's prize valuations and who is informed. If both players pick the same stage, they exert efforts simultaneously. If they pick different stages, the game proceeds sequentially with the early mover as the leader and the late mover as the follower. At stage 1, the players who have picked this stage make their effort at cost c_1. At stage 2, c_2 is revealed to all players, and those who have chosen this stage decide on their effort level after observing any stage 1 efforts. After all efforts have been made, the Tullock mechanism determines the contest winner who receives the prize.

To facilitate the analysis, let us denote the informed strong [weak] player as A [B] and the uninformed strong [weak] player as a [b]. We have two information cases: (i) the strong player is uninformed and the weak player is informed; (ii) the strong player is informed and the weak player is uninformed.

While the complete game admits multiple equilibria, including the standard outcome where the weak player moves first (Baik & Shogren, 1992; Leininger, 1993), we focus our analysis on characterizing equilibria where the strong player moves first. This focus is motivated by the fact that such equilibria represent a reversal of the well-established result in the literature and provide new insights into how cost uncertainty affects strategic behavior in contests. These strong-first equilibria are particularly relevant for applications where stronger players might leverage their market position despite facing uncertainty, such as dominant firms in R&D races or major interest groups in lobbying competitions. Next, we analyze all possible contests that can arise under the two information settings.

3 Equilibrium analysis of subgames

We now analyze the equilibrium outcomes for all possible timing configurations. There are three possible sequences of moves: (*i*) the strong player moves early and the weak player late; (*ii*) the weak player moves early and the strong player late; (*iii*) both players move at the same stage (either early or late). We examine each configuration under both information settings.

3.1 Sequential play with uninformed leader

We first consider the case where the uninformed player has chosen stage 1 and the informed player has chosen stage 2. By observing the informed player's timing decision, the uninformed player might update her beliefs regarding the future cost. This creates a signaling game that may yield multiple equilibria. We examine two types of equilibria: pooling equilibria, where no information is revealed through timing choices, and separating equilibria, where the uninformed player learns the future cost from the informed player's timing decision.

3.1.1 Strong uninformed vs weak informed (aB)

Consider first the case where the strong player lacks information about the future cost. Player a makes effort $x_{a,aB}$ at stage 1, and player B observes this effort and responds with $x_{B,aB}$ at stage 2. Player B's maximization problem is

$$\max_{x_{B,aB}\geq 0}\left\{\frac{x_{B,aB}}{x_{a,aB}+x_{B,aB}}v - c_2 x_{B,aB}\right\}, \quad c_2 \in \{k,1\}\,. \tag{1}$$

At stage 1, player a makes an effort anticipating her opponent's reply. We analyze both pooling and separating equilibria.

Pooling equilibria. In the pooling equilibria, player a does not update her belief about the future cost when observing that player B has chosen stage 2. She maintains her prior belief that $c_2 = k$ with probability q and $c_2 = 1$ with probability $1-q$. Player a's

optimization problem is

$$\max_{x_{a,aB}\geq 0}\left\{q\left[\frac{x_{a,aB}}{x_{a,aB}+x_{B,aB}\left(x_{a,aB}|c_2=k\right)}V-c_1x_{a,aB}\right]\right.$$
$$\left.+\left(1-q\right)\left[\frac{x_{a,aB}}{x_{a,aB}+x_{B,aB}\left(x_{a,aB}|c_2=1\right)}V-c_1x_{a,aB}\right]\right\},\quad c_1\in\{k,1\} \tag{2}$$

where $x_{B,aB}\left(x_{a,aB}|c_2\right)$ denotes player B's optimal response given effort cost c_2.

Lemma 1. *In the pooling equilibrium of the aB contest, the efforts $x_{a,aB}$ and $x_{B,aB}$ exerted by players a and B, respectively, are given by the following:*

- *If $V<2v$,*

$$\left(x^p_{a,aB},\, x^p_{B,aB}\right),\quad \textit{if}\begin{cases} c_1=1,\ \forall c_2 \\ c_1=k,\ c_2\in\{k,\,1\},\ \textit{where } k>k'' \end{cases}$$

$$\left(v,\,0\right),\quad \textit{if } c_1=k,\ c_2\in\{k,\,1\},\ \textit{where } k\leq k'$$

- *If $V\geq 2v$,*

$$\left(x^p_{a,aB},\, x^p_{B,aB}\right),\quad \textit{if}\begin{cases} c_1=1,\ c_2\in\{k,\,1\},\ \textit{where } k<\underline{k} \textit{ and } q>\frac{V-2v}{V} \\ V=2v,\ \forall c_1,c_2 \end{cases}$$

$$\left(\frac{v}{c_2},\,0\right),\quad \textit{if}\begin{cases} c_1=1,\ c_2\in\{k,\,1\},\ \textit{where } k\geq\bar{k} \\ c_1=k,\ \forall c_2 \end{cases}$$

where

$$x^p_{a,aB}=\frac{\left[1-\left(1-\sqrt{k}\right)q\right]^2V^2}{4c_1^2v},$$
$$x^p_{B,aB}=\begin{cases} \frac{[1-(1-\sqrt{k})q]V(2c_1v-kV)}{4c_1^2kv}, & \textit{if } c_2=k \\ \frac{[1-(1-\sqrt{k})q]V(2c_1v-V)}{4c_1^2v}, & \textit{if } c_2=1 \end{cases}$$

and

$$k' = \frac{(1-q)^2 V^2}{(2v - qV)^2}, \quad k'' = \frac{V\left[q^2 V + q\sqrt{V\left[q^2 V + 8(1-q)v\right]} + 4(1-q)v\right]}{8v^2}$$

$$\underline{k} = \frac{\left[2v - (1-q)V\right]^2}{q^2 V^2}, \quad \bar{k} = \frac{(1-q)^2 V + 4qv - (1-q)\sqrt{\left[(1-q)^2 V + 8qv\right]V}}{2q^2 V}.$$

The two players' equilibrium payoffs are

$$\left(\pi^p_{a,aB},\, \pi^p_{B,aB}\right), \quad \text{if } (x_{a,aB},\, x_{B,aB}) = \left(x^p_{a,aB},\, x^p_{B,aB}\right)$$

$$\left(V - c_1 v,\, 0\right), \quad \text{if } (x_{a,aB},\, x_{B,aB}) = \left(\frac{v}{c_2},\, 0\right)$$

where

$$\pi^p_{a,aB} = \frac{\left[1 - \left(1 - \sqrt{k}\right)q\right]^2 V^2}{4c_1 v} \quad \text{and} \quad \pi^p_{B,aB} = \frac{(2c_1 v - c_2 V)^2}{4c_1^2 v}.$$

We observe that when the strong player is not too strong, player B participates in the contest whenever the present cost is high. However, if the present cost is low, player B participates only if this low cost is sufficiently high. Otherwise, the strong leader preempts the weak follower by making the minimum effort that induces zero response given the realized future cost. When the strong player is sufficiently strong, more stringent conditions are required for player B's participation. With a high present cost, player B participates only if the future low cost is sufficiently low and player a believes this occurs with high probability. Intuitively, player a's optimism about future low costs discourages her from exerting high effort, while it encourages player B's participation.

Separating equilibria. In separating equilibria, player a infers from player B's timing choice that the future cost must be low. Otherwise, player B would not concede the first-mover advantage. This creates a sequential game with complete information where player a knows that $c_2 = k$. The following result emerges.

Lemma 2. *In the separating equilibrium of the aB contest, the efforts $x_{a,aB}$ and $x_{B,aB}$ exerted by players a and B, respectively, are given by the following:*

- *If* $V < 2v$,

$$\left(x^s_{a,aB},\, x^s_{B,aB}\right) = \left(\frac{kV^2}{4c_1^2 v},\, \frac{V\left(2c_1 v - kV\right)}{4c_1^2 v}\right), \quad \forall c_1, k$$

- *If* $V \geq 2v$,

$$\left(x^s_{a,aB},\, x^s_{B,aB}\right),\ \textit{if}\ c_1 = 1\ \textit{and}\ k < \frac{2v}{V}$$

$$\left(\frac{v}{k},\, 0\right), \quad \textit{if} \begin{cases} c_1 = 1,\, k \geq \frac{2v}{V} \\ c_1 = k,\, \forall k \end{cases}$$

The separating equilibrium shows that when player a learns that the future cost is low, the conditions for player B's participation become more transparent. If the strong player is not too strong, player B always participates. However, if the strong player is sufficiently strong, player B participates only when both the present cost is high and the future cost is sufficiently low.

3.1.2 Weak uninformed vs strong informed (bA)

The analysis of the bA contest follows similar logic. We present the key results:

Lemma 3. *In the pooling equilibrium of the bA contest, the efforts $x_{b,bA}$ and $x_{A,bA}$ exerted by players b and A, respectively, are*

$$\left(x^p_{b,bA},\, x^p_{A,bA}\right), \quad \textit{if} \begin{cases} c_1 = 1,\, \forall c_2 \\ c_1 = k,\, c_2 \in \{k, 1\},\ \textit{where}\ k > \hat{k} \end{cases}$$

$$(V,\, 0), \quad \textit{if}\ c_1 = k,\, c_2 \in \{k, 1\},\ \textit{where}\ k \leq \hat{k}$$

where

$$x^p_{b,bA} = \frac{\left[1-\left(1-\sqrt{k}\right)q\right]^2 v^2}{4c_1^2 V},$$

$$x^p_{A,bA} = \begin{cases} \frac{[1-(1-\sqrt{k})q]v(2c_1V-kv)}{4c_1^2kV}, & \text{if } c_2 = k \\ \frac{[1-(1-\sqrt{k})q]v(2c_1V-v)}{4c_1^2V}, & \text{if } c_2 = 1 \end{cases}$$

and

$$\hat{k} = \frac{(1-q)^2 v^2}{(2V-qv)^2}.$$

The two players' equilibrium payoffs are

$$\left(\pi^p_{b,bA},\, \pi^p_{A,bA}\right), \quad \text{if } (x_{b,bA},\, x_{A,bA}) = \left(x^p_{b,bA},\, x^p_{A,bA}\right)$$
$$(v - c_1V,\, 0), \quad \text{if } (x_{b,bA},\, x_{A,bA}) = (V,\, 0)$$

where

$$\pi^p_{b,bA} = \frac{\left[1-\left(1-\sqrt{k}\right)q\right]^2 v^2}{4c_1V} \quad \text{and} \quad \pi^p_{A,bA} = \frac{(2c_1V - c_2v)^2}{4c_1^2V}.$$

We notice that in the bA case, the strong follower is preempted only when the present cost is sufficiently low, regardless of her information advantage. Regarding the separating equilibrium, we have the following result.

Lemma 4. *In the separating equilibrium of the bA contest, the efforts $x_{b,bA}$ and $x_{A,bA}$ exerted by players b and A, respectively, are*

$$\left(x^s_{b,bA},\, x^s_{A,bA}\right) = \left(\frac{kv^2}{4c_1^2V},\, \frac{v\,(2c_1V - kv)}{4c_1^2V}\right), \quad \forall c_1, k.$$

In contrast to the aB case, now, the informed strong follower always participates, leveraging both her strength and information advantages.

3.2 Sequential play with informed leader

We now examine the case where the informed player moves at stage 1 and the uninformed player moves at stage 2. In this scenario, the leader knows the exact future cost level.

While the informed player's timing choice could signal information about c_2, this does not affect the follower's decision since c_2 is revealed before the follower chooses her effort. Thus, these are sequential games with complete information by the time efforts are made.

3.2.1 Weak informed vs strong uninformed (Ba)

When the informed weak player leads and the uninformed strong player follows, player B makes effort $x_{B,Ba}$ at stage 1 knowing both c_1 and c_2. Player a observes this effort and, after learning c_2, responds with effort $x_{a,Ba}$. Player a's maximization problem at stage 2 is

$$\max_{x_{a,Ba}\geq 0}\left\{\frac{x_{a,Ba}}{x_{B,Ba}+x_{a,Ba}}V - c_2 x_{a,Ba}\right\}, \quad c_2 \in \{k,1\}.$$

Player B's problem at stage 1, anticipating player a's response, is

$$\max_{x_{B,Ba}\geq 0}\left\{\frac{x_{B,Ba}}{x_{B,Ba}+x_{a,Ba}(x_{B,Ba})}v - c_1 x_{B,Ba}\right\}, \quad c_1 \in \{k,1\}.$$

Lemma 5. *In the Ba contest, the efforts $x_{B,Ba}$ and $x_{a,Ba}$ exerted by players B and a, respectively, are*

$$\left(x^*_{B,Ba},\, x^*_{a,Ba}\right), \quad \text{if} \begin{cases} c_1 = 1,\ \forall c_2 \\ c_1 = c_2 = k \\ c_1 = k,\ c_2 = 1,\ \text{where } k > \frac{v}{2V} \end{cases}$$

$$(V,\,0), \quad \text{if } c_1 = k,\ c_2 = 1,\ \text{where } k \leq \frac{v}{2V},$$

where $x^*_{B,Ba} = \frac{c_2 v^2}{4c_1^2 V}$ and $x^*_{a,Ba} = \frac{v(2c_1V - c_2 v)}{4c_1^2 V}$.
The two players' equilibrium payoffs are

$$\left(\pi^*_{B,Ba},\, \pi^*_{a,Ba}\right), \quad \text{if } (x_{B,Ba},\, x_{a,Ba}) = \left(x^*_{B,Ba},\, x^*_{a,Ba}\right)$$
$$(v - kV,\,0), \quad \text{if } (x_{b,bA},\, x_{A,bA}) = (V,\,0)$$

where

$$\pi^*_{B,Ba} = \frac{c_2 v^2}{4c_1 V} \quad \text{and} \quad \pi^*_{a,Ba} = \frac{(c_2 v - 2c_1 V)^2}{4c_1^2 V}.$$

We observe that the uninformed strong follower can be preempted when the present cost is sufficiently low and the future cost is high. This occurs because player B, knowing that $c_2 = 1$, can exploit the low present cost to make a preemptive effort. This contrasts sharply with the bA case, where the informed strong follower is never preempted.

3.2.2 Strong informed vs weak uninformed (Ab)

In the last possible sequential case, the informed strong player leads, and the uninformed weak player follows.

Lemma 6. *In the Ab contest, the efforts $x_{A,Ab}$ and $x_{b,Ab}$ exerted by players A and b, respectively, are given by the following:*

- *If $V < 2v$,*

$$\left(x^*_{A,Ab},\, x^*_{b,Ab}\right), \quad \text{if} \begin{cases} c_1 = 1,\ \forall c_2 \\ c_1 = c_2 = k \\ c_1 = k,\ c_2 = 1,\ \text{where } k > \frac{V}{2v} \end{cases}$$

$$\left(\frac{v}{c_2}, 0\right), \quad \text{if } c_1 = k,\ c_2 = 1 \text{ where } k \leq \frac{V}{2v}$$

- *If $V \geq 2v$,*

$$\left(x^*_{A,Ab},\, x^*_{b,Ab}\right), \quad \text{if } c_1 = 1,\ c_2 = k \text{ where } k < \frac{2v}{V}$$

$$\left(\frac{v}{c_2}, 0\right), \quad \text{if} \begin{cases} c_1 = k,\ \forall c_2 \\ c_1 = 1,\ c_2 \in \{k, 1\},\ \text{where } k \geq \frac{2v}{V} \end{cases}$$

where $x^*_{A,Ab} = \frac{c_2 V^2}{4c_1^2 v}$ and $x^*_{b,Ab} = \frac{V(2c_1 v - c_2 V)}{4c_1^2 v}$.

When the players are not sufficiently asymmetric, the weak follower abstains only when facing unfavorable cost conditions, i.e., low present cost and high future cost. However, when the strong player is sufficiently strong, the weak follower participates only under very favorable conditions, i.e., high present cost and sufficiently low future cost.

3.3 Simultaneous play

Finally, we consider the case where both players choose the same stage to exert effort. When both players move at stage 1, they both know the present cost, c_1. When both move at stage 2, the uninformed player learns c_2 before choosing her effort. Thus, in both cases, the players compete under complete and symmetric information.

Player i's maximization problem is

$$\max_{x_i \geq 0} \left\{ p_i \left(x_i, x_j \right) v_i - d x_i \right\},$$

where $v_i \in \{V, v\}$ is player i's prize valuation and $d \in \{k, 1\}$ is the common cost at the time of competition.

In equilibrium, player i's effort is

$$x^*_{i,S} = \frac{v_i^2 v_j}{d \left(v_i + v_j \right)^2}$$

and her payoff is

$$\pi^*_{i,S} = \frac{v_i^3}{\left(v_i + v_j \right)^2}.$$

Note that these payoffs are independent of d, as both players face the same cost. This complete information benchmark will be important when analyzing the timing decisions in Section 5.

4 Expected payoffs before timing decisions

Before analyzing the endogenous timing equilibrium, we must determine the two players' expected payoffs at stage zero when they decide whether to move early or late. For uninformed leaders, these payoffs are already given by the equilibrium payoffs in Section 3. However, when an uninformed player considers moving late while the informed player moves early, she faces uncertainty about c_2 when making her timing decision.

Assume that the uninformed player, i, chooses stage 2 while the informed player, J,

chooses stage 1. At stage zero, player i's expected payoff is

$$\Pi_{i,Ji} = q\pi^*_{i,Ji}(c_2 = k) + (1-q)\,\pi^*_{i,Ji}(c_2 = 1)$$

where $\pi^*_{i,Ji}(c_2)$ denotes player i's equilibrium payoff when the future cost is c_2.

The following lemma provides these expected payoffs for uninformed followers:

Lemma 7. *At stage zero of the Ba contest, player a's expected payoff is*

$$\Pi_{a,Ba} = \begin{cases} \frac{q(2V-v)^2}{4V}, & \text{if } k \leq \frac{v}{2V} \\ V + \frac{\left[1-\left(1-k^2\right)q\right]v^2}{4c_1^2 V} + \frac{[q(1-k)-1]v}{c_1}, & \text{otherwise.} \end{cases}$$

At stage zero of the Ab contest, player b's expected payoff is given by the following:

- *If $V < 2v$,*

$$\Pi_{b,Ab} = \begin{cases} \frac{q(V-2v)^2}{4v}, & \text{if } V < 2v \text{ and } c_1 = k \\ v + \frac{\left[1-\left(1-k^2\right)q\right]V^2}{4c_1^2 v} + \frac{[q(1-k)-1]V}{c_1}, & \text{otherwise.} \end{cases}$$

- *If $V \geq 2v$,*

$$\Pi_{b,Ab} = \begin{cases} \frac{q(kV-2v)^2}{4v}, & \text{if } c_1 = 1 \text{ and } k < \frac{2v}{V} \\ 0, & \text{otherwise.} \end{cases}$$

With these expected payoffs established, we can proceed with analyzing the players' timing decisions.

5 Timing Equilibrium

The two players simultaneously choose whether to move early (stage 1) or late (stage 2). Table 1 illustrates the normal form of the extended game after replacing the sub-games with their solutions.[7] For convenience, we assume that if the players are indifferent

[7]Keep in mind that Table 1 is not unique, but multiple tables exist depending on the two players' efforts and the corresponding levels of equilibrium payoffs as well as on the type of equilib-

between moving early or late, they choose to do the former.

Player i \ Player J	Stage 1	Stage 2
Stage 1	$\Pi_{i,S}$, $\Pi_{J,S}$	$\Pi_{i,iJ}$, $\Pi_{J,iJ}$
Stage 2	$\Pi_{i,Ji}$, $\Pi_{J,Ji}$	$\Pi_{i,S}$, $\Pi_{J,S}$

Table 1: Player i and player J's expected payoffs.

A timing equilibrium specifies each player's stage choice such that neither player wants to deviate given the other's choice. While multiple equilibria with the standard weak-first equilibrium established in the literature may exist, we focus on characterizing when the strong player moves first.

5.1 Uninformed Strong, Informed Weak

We first examine when player a (uninformed strong) moves early while player B (informed weak) moves late. For this to be an equilibrium, it must be that $\Pi_{a,aB} \geq \Pi_{a,S}$, i.e., player a prefers to lead, and $\Pi_{B,aB} \geq \Pi_{B,S}$, i.e., player B prefers to follow. The following proposition characterizes when these inequalities are satisfied.

Proposition 1. *A timing equilibrium where the uninformed strong player leads and the informed weak player follows (aB) exists if and only if $c_1 = 1$ and $c_2 = k$, and the following conditions are satisfied:*

- *If B is sufficiently strong, then q and k are both low or both moderate*
- *If B is moderately strong, then q is low for every k, or q is moderate and k is low, or q is high and k is moderate*
- *If B is moderately weak, then q is low for every k, or q is moderate and k is high, or q is high and k is moderate*

rium played. For instance, for the case where player i moves first, we have that $(\Pi_{i,iJ}, \Pi_{J,iJ}) \in \left\{\left(\pi^p_{i,iJ}, \pi^p_{J,iJ}\right), \left(\pi^s_{i,iJ}, \pi^s_{i,iJ}\right), (V - c_1 v, 0), \left(V - \frac{c_1 v}{k}, 0\right), (v - c_1 V, 0)\right\}$.

If B sufficiently weak, a timing equilibrium aB does not exist.

Proposition 1 demonstrates that if the present cost is high and the future one is low, there are several conditions under which player a prefers to move early and player B follows. First, if player B is sufficiently strong, then player a should believe that the future cost will be low with a low probability or moderate with a moderate probability. In such a case, player B, having perfect information about the future low cost, chooses to move late and pay a low cost instead of a greater one if she moves early. However, player a is not fully informed about the future cost but only knows that it is not likely to be low or it is possible to be moderate. Therefore, she chooses to move early. It is interesting that although player a is not much stronger than player B, she still prefers to submit an effort at stage 1, although she expects that the future cost will be lower than the present one

As B becomes weaker, a low probability that the future cost is low is adequate for player a to choose to exert effort at stage 1. Moreover, player B, being weaker now, seeks the low future cost to mitigate her opponent's advantage. Similarly, in the case where player a believes that the future cost is moderate with a high probability, the effect of the first-mover advantage dominates the one of paying a lower cost, and, consequently, player a makes an effort at stage 1. However, if player a believes that the future cost is lower with a moderate probability, whether a timing equilibrium aB exists depends on the two players' strengths and the future cost level. Specifically, if player B is moderately strong, the future cost should be low to induce her to make an effort at stage 2. On the contrary, as player B becomes weaker, even if the future cost is high, she still prefers to exert effort at stage 2 and pay that cost as it is lower than the present one. By doing so, player B strives to reduce the two players' strength differences as much as possible. Finally, as player B becomes even weaker, she is preempted by the strong leader.

Investigating the two players' timing choices under the remaining cost combinations yields the following result.

Corollary 1.1. *For any $(c_1, c_2) \neq (1, k)$, a timing equilibrium where the uninformed strong player leads does not exist.*

This corollary emphasizes that the reversal of standard timing requires specific cost

conditions, i.e., high present costs and low future costs, with only the weak player knowing this favorable future.

5.2 Informed Strong, Uninformed Weak

We now examine when player A (informed strong) moves early while player b (uninformed weak) moves late. We obtain the following result.

Proposition 2. *A timing equilibrium where the informed strong player leads and the uninformed weak player follows (Ab) exists if and only if* $c_1 = c_2 = 1$*, and both* $q > \hat{q}$ *and* $k < \frac{2v}{V+v}$.

Proposition 2 indicates that if both present and future costs are high, for a timing equilibrium Ab to exist, player b should expect that the future cost will be sufficiently low with a high probability. Assume instead that the present cost is low. In such a case, although player A finds it optimal to pick stage 1 for her move, player b also prefers to choose that stage. Now, assume that the present cost is high and the future cost is low. In this case, player A chooses to make an effort at stage 1 only if the future cost is sufficiently high, but player b chooses to make an effort at stage 2 only if she believes that the future cost might be low with a high probability.

The following corollary summarizes the players' timing choices under all the possible cost combinations discussed above.

Corollary 2.1. *For any* $(c_1, c_2) \neq (1, 1)$*, a timing equilibrium where the informed strong player leads does not exist.*

5.3 Comparison with Standard Results

Our results contrast sharply with the standard finding that weak players move first in endogenous timing contests. In the standard models with complete information, the weak player moves first to soften competition. The principal difference in our model is the combination of cost uncertainty and information asymmetry, which results in a reversed order of moves. Particularly, when the strong player is uninformed about future costs, she moves first because the value of preemption outweighs the option value of

waiting, especially when she is pessimistic about the future state. If the strong player possesses full information, she moves first to exploit her knowledge that waiting offers no cost advantage, while the weak player's optimism about future costs induces her to wait.

6 Optimal Information Disclosure

In the previous analysis, we demonstrated that an order of moves with the strong player moving first occurs only under specific beliefs. In this section, we explore how a contest designer who maximizes total effort can strategically manipulate information disclosure to induce her optimal timing of efforts.

Consider a designer who can choose between two instruments: price discrimination or information disclosure. Under price discrimination with complete information, the designer offers a lower price to early movers. From Protopappas (2023), we know that when the designer offers a lower price to early movers under complete information, both players move early in equilibrium. The resulting total effort is $E^*_{\text{price}} = \frac{Vv}{V+v}$, i.e, the total effort in a simultaneous contest with effort cost equal to 1 (high cost) for both players.

Now, suppose the designer cannot discriminate on price but can control information. The designer's strategy is characterized by $I = (q_A, q_b)$, where q_i is player i's belief that $c_2 = k$ when uninformed. The designer can induce the Ab timing pattern by setting $q_A = 0$, i.e., fully informing the strong player that $c_2 = 1$, or setting $q_b > \hat{q}$, i.e., making the weak player optimistic about future costs. Under this information structure, player A moves first, knowing that the cost will remain high, while player b follows, hoping for a lower future cost. The resulting total effort is given by the following.

Lemma 8. *Under the optimal information disclosure policy, $I = (0, q_b)$ with $q_b > \hat{q}$ and $k < \frac{2v}{V+v}$, total effort is*

$$E^*_{info} = \frac{Vv}{V+v} + \frac{v(V-v)}{2(V+v)^2} > E^*_{price}$$

The second term of E^*_{info} represents the additional effort generated by sequential play with the strong player leading. This occurs because, first, the strong leader exerts higher effort than in simultaneous play, and also, the weak follower, despite hoping for low costs

that don't materialize, still participates. We summarize in the following result.

Proposition 3. *When $c_1 = c_2 = 1$ and $k < \frac{2v}{V+v}$, information disclosure strictly dominates price discrimination for an effort-maximizing contest designer. The optimal policy fully informs the strong player about high future costs, while maintaining the weak player's optimism about cost reductions.*

The above result has an important implication. Rather than using monetary incentives, contest designers can use information as a tool to achieve superior outcomes. This is especially relevant when price discrimination is legally constrained or practically difficult.

7 Conclusion

We have studied an asymmetric Tullock contest where two players can exert effort early or late, and there is uncertainty regarding the future effort cost. We have demonstrated that, if the present cost is low, even an uninformed weak leader might preempt a strong informed follower. Introducing endogenous timing of moves, we have shown that there are equilibria in which the strong player—informed or uninformed—moves early and the weak player follows, demonstrating that the common equilibria with the weak player moving early and the strong player following or a simultaneous move by the two players are not the only possible outcomes.

The main contribution of the study is identifying conditions under which cost uncertainty reverses the standard timing result. When the uninformed strong player is sufficiently pessimistic about future costs and the informed weak player knows costs will be favorable, the strong player's desire for first-mover advantage dominates the option value of waiting. Similarly, when the informed strong player knows that the cost will remain high and the uninformed weak player is optimistic, sequential play with the strong player leading emerges in equilibrium. We have also shown that an effort-maximizing contest designer might find it optimal not to affect the effort prices she charges but, instead, the information she discloses to the players regarding the future cost. By strategically creating information asymmetry, i.e., fully informing the strong player while maintaining the weak player's optimism, the designer can induce effort levels that exceed those achievable through price discrimination.

An extension of the model would be the dependence of the future effort cost on the effort level of the early mover. For instance, in an innovation contest, a high early mover investment might drive future investment prices up. This task is left for future research.

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A Proofs

Proof of Lemma 1. We start with Player B who chooses her effort taking into account Player a's effort who has moved at stage 1. The FOC of (1) is

$$\frac{x_{a,aB}}{(x_{a,aB}+x_{B,aB})^2}v - c_2 = 0$$

which yields the reaction function $x_{B,aB}(x_{a,aB}) = \sqrt{\frac{x_{a,aB}v}{c_2}} - x_{a,aB}$. Following the analysis in Leininger (1993), since it must be $x_{B,aB} \geq 0$, i.e., the effort cannot be negative, Player B makes effort

$$x_{B,aB}(x_{a,aB}) = \begin{cases} \sqrt{\frac{x_{a,aB}v}{c_2}} - x_{a,aB}, & \text{if } x_{a,aB} < \frac{v}{c_2} \\ 0, & \text{if } x_{a,aB} \geq \frac{v}{c_2}. \end{cases} \tag{3}$$

Player a chooses her level of effort to exert, anticipating player B's optimal response. In the pooling equilibrium, player a believes that with probability q, player B will face cost k and with probability $1-q$, player B will face cost 1. Substituting player B's reaction function into (2), we can express player a's expected payoff function as

$$\Pi_{a,aB}(x_{a,aB}) = \begin{cases} \frac{[1-(1-\sqrt{k})q]V\sqrt{x_{a,aB}}}{\sqrt{v}} - c_1 x_{a,aB}, & \text{if } x_{a,aB} < \min\left\{\frac{v}{k}, v\right\} \\ V - c_1 x_{a,aB}, & \text{if } x_{a,aB} \geq \frac{v}{c_2} \text{ for all possible } c_2. \end{cases}$$

For $x_{a,aB} < \min\{v/k, v\} = v/\max\{k, 1\}$, the FOC of player a's maximization problem is

$$\frac{[1-(1-\sqrt{k})q]V}{2\sqrt{x_{a,aB}v}} - c_1 = 0.$$

Solving for $x_{a,aB}$, we find that player a exerts effort $x^p_{a,aB} = \frac{[1-(1-\sqrt{k})q]^2V^2}{4c_1^2 v}$. Substituting $x^p_{a,aB}$ into (3), player B replies with an effort that depends on the realized c_2. For $x_{a,aB} \geq \frac{v}{c_2}$, player a's payoff function is decreasing in $x_{a,aB}$ and, thus, it is optimal for her to exert the minimum possible effort that induces zero response from player B for all possible realizations of c_2, i.e., v.

We consider two cases: $V < 2v$ and $V \geq 2v$ and check the players' behavior under all

possible combinations of c_1 and c_2.

In the first case, $V < 2v$, we have the following: for $c_1 = c_2 = 1$ and $c_1 = c_2 = k$, it is always true that $x^p_{a,aB} < \frac{v}{c_2}$ and, therefore, players exert efforts $x^p_{a,aB}$ and $x^p_{B,aB}$. For $c_1 = 1, c_2 = k$, we have $x^p_{a,aB} < \frac{v}{k} \iff k > \frac{(1-q)^2V^2}{(2v-qV)^2} \equiv k'$. For $c_1 = k, c_2 = 1$, we have $x^p_{a,aB} < v \iff k > \frac{V[q^2V+q\sqrt{V[q^2V+8(1-q)v]}+4(1-q)v]}{8v^2} \equiv k''$. Since $k' < k''$ for all valid parameters, if $k > k''$, both types of player B participate. If $k \leq k'$, player a preempts both types of player B by making effort v.[8]

In the second case, $V \geq 2v$, we have: For $c_1 = c_2 = 1$, $x^p_{a,aB} < v$ if $V = 2v$ or if $q > \frac{V-2v}{V}$ and $k < \frac{[2v-(1-q)V]^2}{q^2V^2} \equiv \underline{k}$. For $c_1 = 1, c_2 = k$, we have $x^p_{a,aB} < \frac{v}{k} \iff k < \frac{(1-q)^2V+4qv-(1-q)\sqrt{[(1-q)^2V+8qv]V}}{2q^2V} \equiv \bar{k}$. For $c_1 = c_2 = k$ and $c_1 = k, c_2 = 1$, it always holds that $x^p_{a,aB} \geq \frac{v}{c_2}$ implying that player a makes effort $\frac{v}{c_2}$ where c_2 is the realized cost.

Finally, we obtain the two players' equilibrium payoffs by substituting their equilibrium efforts into their payoff functions. □

Proof of Lemmas 2 and 4 to 6. In the separating equilibrium with the uninformed player exerting effort at stage 1 and the informed at stage 2 (Lemmas 2 and 4), the former updates her belief about c_2 when the latter chooses to make an effort late. In the cases with the informed player exerting effort at stage 1 and the uninformed at stage 2, the latter learns the cost before she makes an effort, implying no cost uncertainty in the contest. All the above cases resemble the one with full information described in Protopappas (2023, Lemma 1) when the strong player moves early and the weak moves late, and in Protopappas (2023, Lemma 2) when we have the opposite order of moves. □

Proof of Lemma 7. At stage zero, if player a decides to move at stage 2, there is a probability q that $c_2 = k$ and a probability $1-q$ that $c_2 = 1$. Therefore, with probability q, the two players make efforts $\left(x^*_{B,Ba}, x^*_{a,Ba}\right)$ and with probability $1-q$, they make efforts $(V, 0)$. It follows that, if $c_1 = k \leq \frac{v}{2V}$, player a's expected payoff is

$$\begin{aligned}\Pi_{a,Ba} &= q\frac{(kv-2kV)^2}{4k^2V} + (1-q)\,0 \\ &= \frac{q\,(2V-v)^2}{4V}.\end{aligned}$$

[8] For $k \in (k', k'']$, no pure strategy equilibrium exists in the pooling case.

In any other case, player a's expected payoff is

$$\begin{aligned}\Pi_{a,Ba} &= q\pi^*_{a,Ba}\left(c_2 = k\right) + \left(1-q\right)\pi^*_{a,Ba}\left(c_2 = 1\right)\\ &= q\frac{\left(kv - 2c_1V\right)^2}{4c_1^2V} + \left(1-q\right)\frac{\left(v - 2c_1V\right)^2}{4c_1^2V}\\ &= V + \frac{\left[1-\left(1-k^2\right)q\right]v^2}{4c_1^2V} + \frac{\left[q\left(1-k\right)-1\right]v}{c_1}.\end{aligned}$$

The same logic applies to finding b's expected payoff at stage zero of the Ab contest. □

Proof of Proposition 1. In the following, we compare player a and B's expected payoffs under all the possible move orders and cost combinations. We focus on the timing equilibria with the strong player moving at stage 1 and the weak at stage 2. Therefore, we consider a pooling equilibrium at stages 1 and 2 since in the case of a separating equilibrium, an aB equilibrium does not exist Protopappas (2023, Lemma 3).

Let us begin with the case $v \geq \frac{V}{2}$. For $(c_1, c_2) = (1, 1)$, if player B picks stage 2, we have $\Pi_{a,aB} > \Pi_{a,S}$ if $\left(q < \underline{q}, \forall k\right)$ or $\left(q > \underline{q} \text{ and } k > K\right)$ where $\underline{q} = 1 - 2\sqrt{\frac{Vv}{(V+v)^2}}$ and $K = \frac{2\left(q^2-2q+3\right)Vv+(1-q)^2v^2+(1-q)^2V^2}{q^2(V+v)^2} - 4\sqrt{\frac{(1-q)^2vV}{q^4(V+v)^2}}$. If player a picks stage 1, we have $\Pi_{B,aB} > \Pi_{B,S}$ if $q > \frac{V-v}{V+v}$ and $k < \frac{\left[(1+q)v-(1-q)V\right]^2}{q^2(V+v)^2}$. For the first pair of conditions, we have $\underline{q} < \frac{V-v}{V+v}$, i.e, q can never be simultaneously lower than $\underline{q}$ and greater than $\frac{V-v}{V+v}$. For the second pair of conditions, we have $\underline{q} < \frac{V-v}{V+v}$, i.e., $q > \underline{q}$ and $q > \frac{V-v}{V+v}$ reduce to $q > \frac{V-v}{V+v}$. For $q > \frac{V-v}{V+v}$, we have $K > \frac{\left[(1+q)v-(1-q)V\right]^2}{q^2(V+v)^2}$ which implies that k cannot be simultaneously lower than $\frac{\left[(1+q)v-(1-q)V\right]^2}{q^2(V+v)^2}$ and greater than K. We conclude that aB is not an equilibrium. For $c_1 = c_2 = k > k''$, if player a picks stage 1, we have $\Pi_{B,aB} > \Pi_{B,S}$ if $k < \frac{(1-q)^2V^2(V+v)^2}{\left[qV^2+4v^2+(2-q)Vv\right]^2}$. However, $k'' < \frac{(1-q)^2V^2(V+v)^2}{\left[qV^2+4v^2+(2-q)Vv\right]^2}$, $\forall q$, i.e., aB is not an equilibrium. For $c_1 = c_2 = k < k'$, if player a picks stage 1, we have $\Pi_{B,aB} > \Pi_{B,S}$, $\forall q$ and, therefore, also in this case, Ab is not an equilibrium. For $(c_1, c_2) = (k, 1)$, if player a picks stage 1, in both cases with $k > k''$ and $k < k'$, we have $\Pi_{B,aB} > \Pi_{B,S}$, i.e., player B never picks stage 2 and, therefore, aB is not an equilibrium. For $(c_1, c_2) = (1, k)$, if player B picks stage 2, we have $\Pi_{a,aB} > \Pi_{a,S}$ if $q < \underline{q}$ or $\left(q > \underline{q} \text{ and } k > \underset{\sim}{k}\right)$ where $\underset{\sim}{k} = \frac{q^2(V+v)^2-2q(V+v)^2-4(1-q)\sqrt{Vv}(V+v)+v^2+6Vv+V^2}{q^2(V+v)^2}$. If player a picks stage 1, we have $\Pi_{B,aB} > \Pi_{B,S}$ if $k < \tilde{k}$ where $\tilde{k} = \frac{1}{2}\left[\frac{(1+q)^2v+(1-q)^2V}{q^2(V+v)} - \sqrt{\frac{(q^2+6q+1)(1-q)^2v+(1-q)^4V}{q^4(V+v)}}\right]$. Combining the above restrictions, we get that aB is a timing equilibrium if $\left(q < \underline{q} \text{ and } k < \tilde{k}\right)$

or $\left(\underline{q} < q < \bar{q} \text{ and } \underline{k} < k < \tilde{k}\right)$ where $\bar{q} = \frac{V}{V+v} - \sqrt{\frac{Vv}{(V+v)^2}}$.

Now, consider the case $v < \frac{V}{2}$. For $c_1 = c_2 = 1$ where $k < \underline{k}$ and $q > \frac{V-2v}{V}$, if player B picks stage 2, we have $\Pi_{a,aB} > \Pi_{a,S}$ if $\left(q \leq \underline{q} \text{ and } k < \underline{k}\right)$ or $\left(q > \underline{q} \text{ and } K < k < \underline{k}\right)$. If player a picks stage 1, we have $\Pi_{B,aB} > \Pi_{B,S}$ if $q > \frac{V-v}{V+v}$ and $k < \frac{[(1+q)v-(1-q)V]^2}{q^2(V+v)^2}$. However, $\frac{V-v}{V+v} > \underline{q}$ and, thus, it cannot be that $q > \frac{V-v}{V+v}$ and $q \leq \underline{q}$. Thus, it should be $q > \underline{q}$ for player B to pick stage 2 and, consequently, $q > \frac{V-v}{V+v}$ for player a to pick stage 1. For $q > \frac{V-v}{V+v}$ then, we have $\frac{[(1+q)v-(1-q)V]^2}{q^2(V+v)^2} < K$ which implies that k cannot be both greater than K and lower than $\underline{k}$. We conclude that an Ab equilibrium does not exist. For $c_1 = c_2 = k$ where $k > \bar{k}$, if player a pick stage 1, we have $\Pi_{B,S} > \Pi_{B,aB} \equiv 0$, i.e., player B also picks stage 1 and Ab is not an equilibrium. For $c_1 = c_2 = k$, if player a picks stage 1, both for $k > k''$ and $k < k'$, we have $\Pi_{B,aB} < \Pi_{B,S}$, i.e., player B never picks stage 2. Similarly for $(c_1, c_2) = (k, 1)$. For $(c_1, c_2) = (1, k)$ where $k < \underline{k}$ and $q > \frac{V-2v}{V}$, if player B picks stage 2, we have $\Pi_{a,aB} > \Pi_{a,S}$ if $v > \frac{1}{3}\left[\sqrt[3]{\frac{1}{2}\left(29+3\sqrt{93}\right)}V + \sqrt[3]{\frac{2}{29+3\sqrt{93}}}V - 2V\right]$ and $\left(\frac{V-2v}{V} < q \leq \underline{q} \text{ and } k < \underline{k}\right)$ or $\left(q > \underline{q} \text{ and } K < k < \underline{k}\right)$. If player a moves early, we have two subcases. If $v \leq \frac{(\sqrt{3}-1)V}{2}$, we have $\Pi_{B,aB} > \Pi_{B,S}$ if $q > \frac{V-2v}{V}$ and $k < \underline{k}$ meaning that B always picks stage 2. If $v > \frac{(\sqrt{3}-1)V}{2}$, we have $\Pi_{B,aB} > \Pi_{B,S}$, if $\left(\frac{V-2v}{V} < q \leq \frac{V^2-Vv-2v^2}{Vv} \text{ and } k < \underline{k}\right)$ or $\left(q > \frac{V^2-Vv-2v^2}{Vv} \text{ and } k < \tilde{k}\right)$. We proceed by combining all the above restrictions. First, for $v \leq \frac{1}{3}\left[\sqrt[3]{\frac{1}{2}\left(29+3\sqrt{93}\right)}V + \sqrt[3]{\frac{2}{29+3\sqrt{93}}}V - 2V\right]$, player a never picks stage 1 and, thus, aB is not an equilibrium. For $\frac{1}{3}\left[\sqrt[3]{\frac{1}{2}\left(29+3\sqrt{93}\right)}V + \sqrt[3]{\frac{2}{29+3\sqrt{93}}}V - 2V\right] < v < \sqrt{0.49}V$, aB is an equilibrium if $\left(\frac{V-2v}{V} < q < \underline{q} \text{ and } k < \underline{k}\right)$ or $\left(\underline{q} < q < \frac{V^2-2v^2-Vv}{Vv} \text{ and } K < k < \underline{k}\right)$) or $\left(\frac{V^2-2v^2-Vv}{Vv} < q < \bar{q} \text{ and } K < k < \tilde{k}\right)$. For $\sqrt{0.49}V \leq v < \frac{V}{2}$, aB is an equilibrium if $\left(\frac{V-2v}{V} < q < \frac{V^2-2v^2-Vv}{Vv} \text{ and } k < \underline{k}\right)$ or $\left(\frac{V^2-2v^2-Vv}{Vv} < q < \underline{q} \text{ and } k < \tilde{k}\right)$ or $\left(\underline{q} < q < \bar{q} \text{ and } K < k < \tilde{k}\right)$. □

Proof of Proposition 2. We use Lemma 7 to calculate player b's expected payoffs. We consider only the case $V < 2v$ as we already know from Protopappas (2023) that an Ab equilibrium cannot emerge when $V \geq 2v$. We have then, for $(c_1, c_2) = (k, 1)$, where $k > \frac{V}{2v}$, if player b picks stage 2, we have $\Pi_{A,Ab} > \Pi_{A,S}, \forall q, k$, i.e., player A always picks

stage 1. If player A picks stage 1, we have that $\Pi_{b,Ab} < \Pi b_S$, $\forall V, v, q, k$, i.e., player b also picks stage 1. Thus, Ab is not an equilibrium. Similarly for $(c_1, c_2) = \{(k, 1), (k, k)\}$. For $(c_1, c_2) = (1, k)$, if player b picks stage 2, we have $\Pi_{A,Ab} > \Pi_{A,S}$ if $k > \frac{4Vv}{(V+v)^2}$. If player A picks stage 1, we have $\Pi_{b,Ab} > \Pi_{b,S}$ if $k < \frac{2v}{V+v}$ and $q > \hat{q} \equiv \frac{(V-v)\left(V^2-Vv-4v^2\right)}{(1-k)(V+v)^2[(1+k)V-4v]}$. However, $\frac{4Vv}{(V+v)^2} > \frac{2v}{V+v}$, $\forall q, k$ and, therefore, k cannot be simultaneously greater than $\frac{4Vv}{(V+v)^2}$ and lower than $\frac{2v}{V+v}$. Therefore, Ab is not an equilibrium. For $(c_1, c_2) = (1, 1)$, if player b picks stage 2, we have $\Pi_{A,Ab} > \Pi_{A,S}, \forall q, k$. If player A picks stage 1, we have $\Pi_{b,Ab} > \Pi_{b,S}$ if $k < \frac{2v}{V+v}$ and $q > \hat{q}$. Therefore, there is an Ab equilibrium if these conditions are met.

$\square$